

Block III—Topic 7:
**Owning vs. Renting: Externalities,
Determinants and Policies**



Lent Term Lectures so far...

- Explored role of government in real estate markets
 - ▶ Fiscal differentials and shocks are **capitalized** into house prices and rents
 - ▶ Extent depends on **long-run supply constraints** (incl. **land use restrictions**)
 - Tried to make sense of **house price dynamics** in different markets
 - ▶ **Exogenous vs. endogenous cycles**
- ⇒ **Very different implications for existing homeowners vs. renters...**

This Lecture

1. Some Stylized Facts about Homeownership Attainment
2. Motivation: (Why) Should We Care about Homeownership Attainment?
3. Homeownership Subsidies: Justified or Not? (Market Failure vs. Policy Failure)
4. What Factors Determine Individual Tenure Choice and Local Homeownership Rates?

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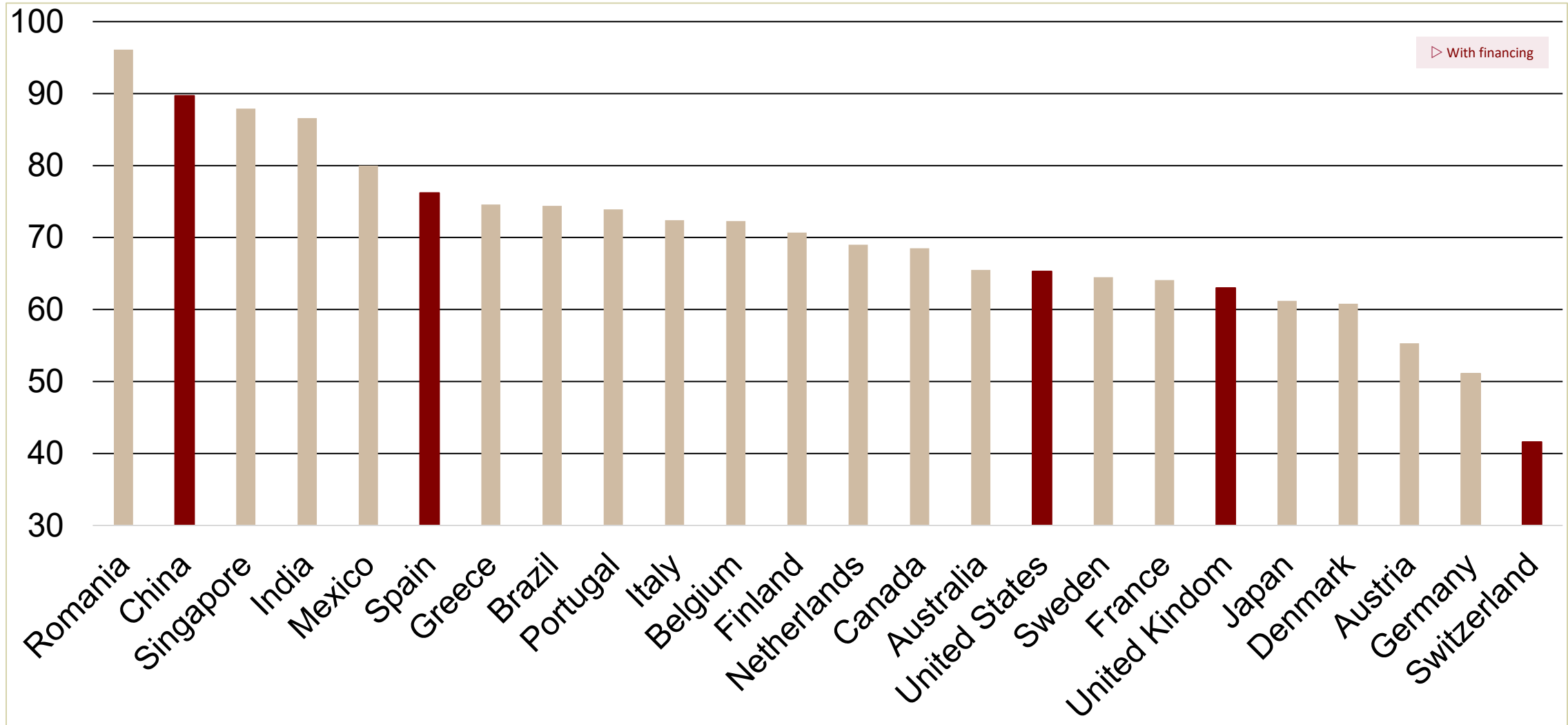
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Stylised Fact 1

- Homeownership rates vary significantly **across countries** and **over time**...



Homeownership Rates — Selected Countries (2020)*



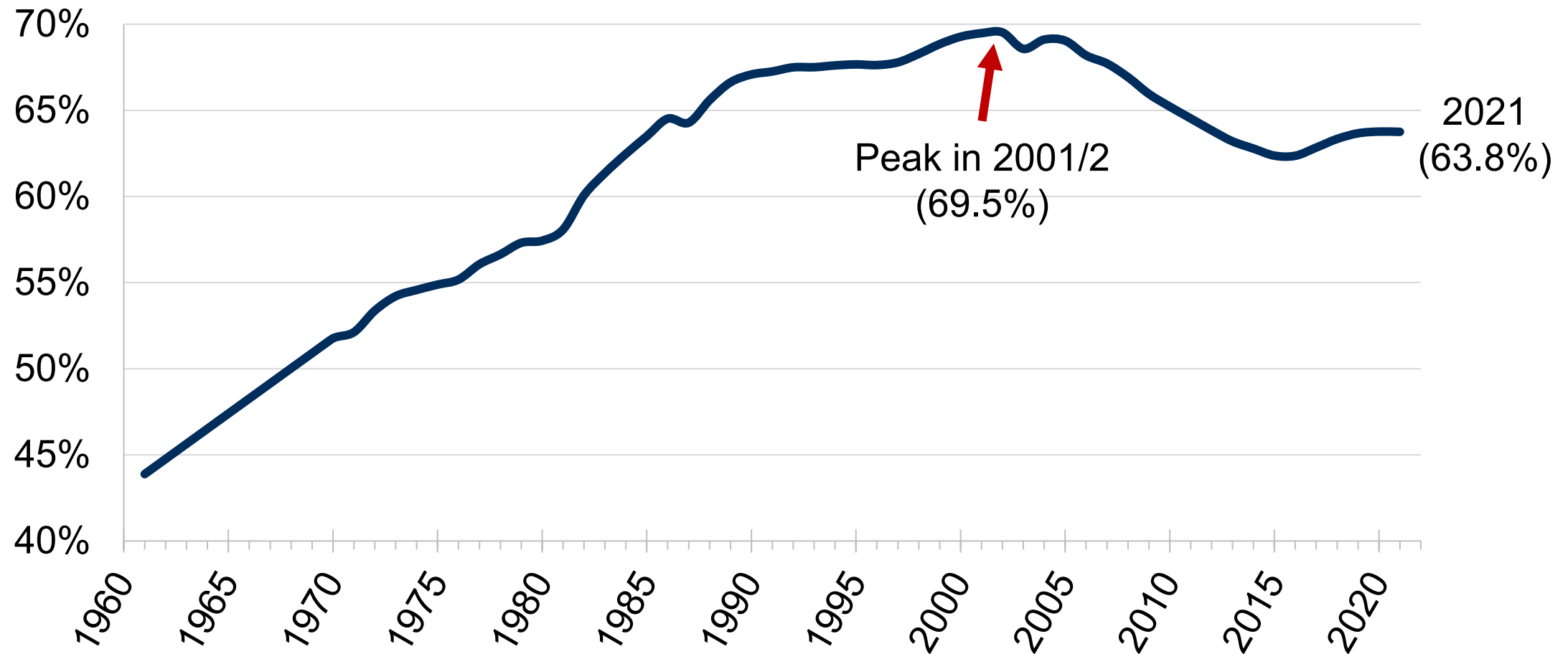
Sources: Various national statistics, *Except India (2011), Mexico and Brazil (2009), Australia (2016) and Japan (2017)

Changes in Homeownership Rates over Longer Time-Periods

Country	1945-50	2000	2020	Change
UK	29%	69%	65%	+36%
Italy	40%	78%	75%	+35%
Australia	53%	69%	66%	+13%
US	57%	67%	63%	+6%
Canada	66%	66%	66%	nil

Sources: Doling (1997, Table. 9.1, 159), Danish National Agency for Enterprise and Housing (2003), ECHP (2000), Australian Bureau of Statistics, US Census Bureau, Canadian Home Builder's Association, English Housing Survey, Statistics Canada, Eurostat, United Nations

UK Homeownership Rate, 1961-2021

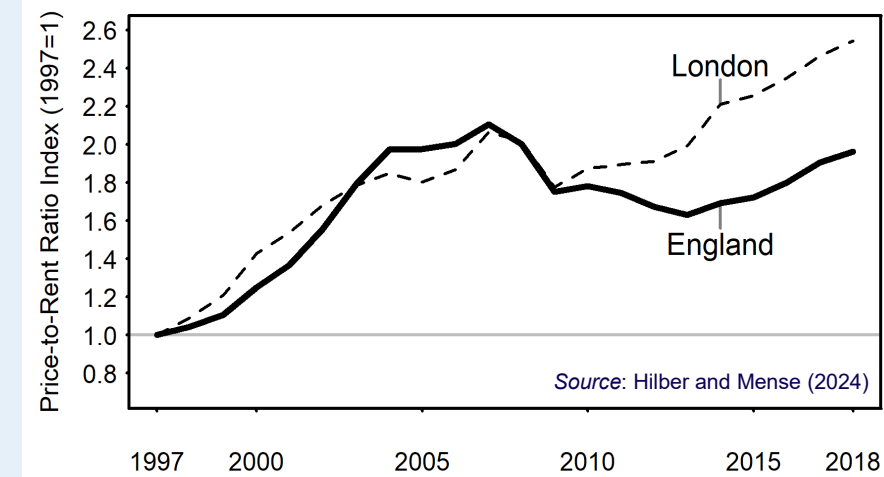


Source: MHCLG, Live Table 104 (downloaded January 2023)

⇒ Possible reason(s) for decline since 2001/2? **Discuss...**

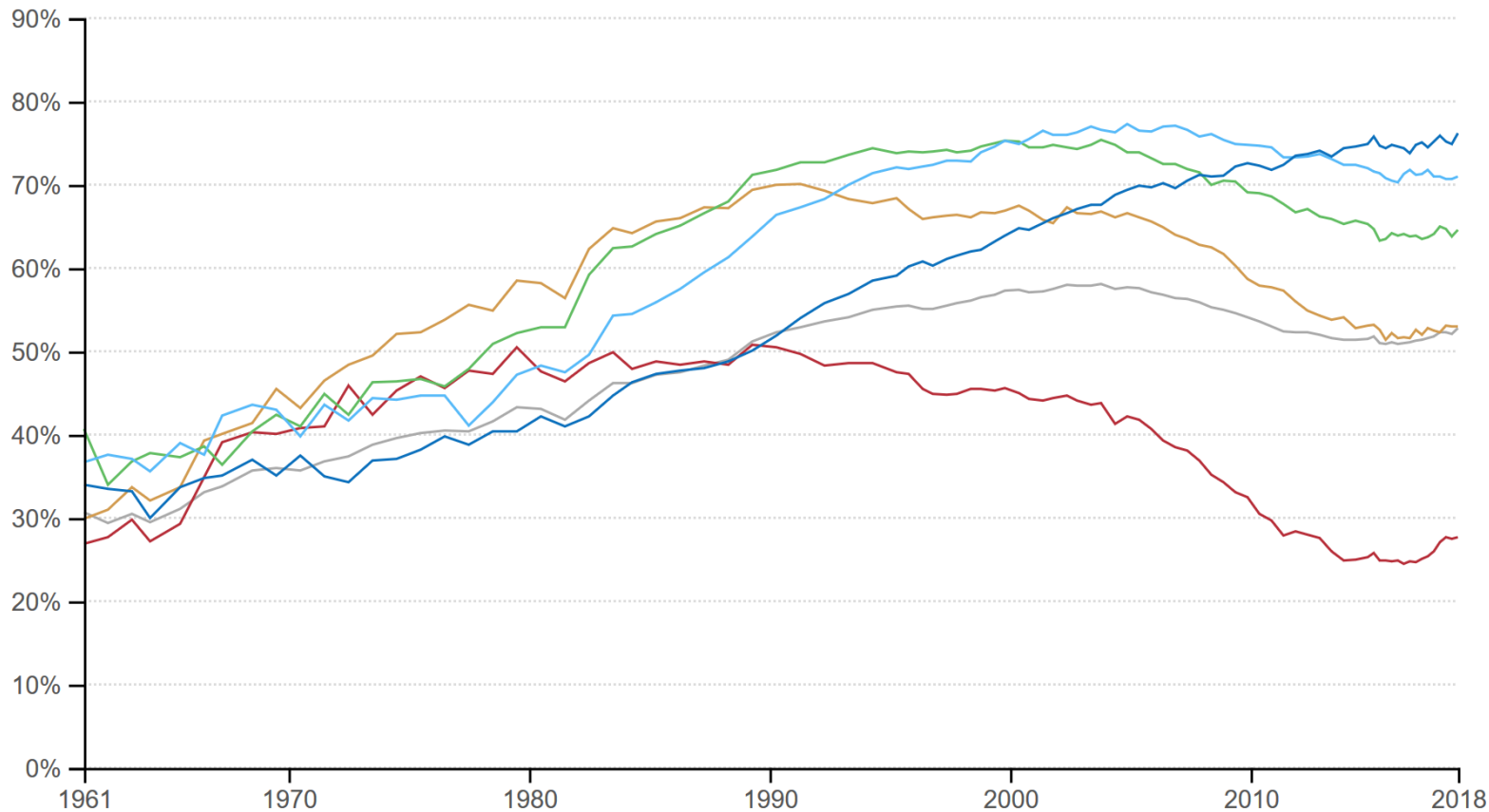
Possible explanations for decline in homeownership rate since 2001/2 (*Discuss*)

- Increase in price-to-rent ratio
 - But increase started before 2001/2
- Changing preferences of younger households?
- High prices in conjunction with down-payment- and liquidity-constraints
 - Lower income HHs & young increasingly face serious constraints
⇒ Would like to buy but increasingly can't!
- Why recovery from 2016?
 - Possibly due to Help-to-Buy?
(But introduced in 2013!)



UK Homeownership Attainment Across Age Cohorts, 1961-2018

% of families (singles or couples) owning their own home: UK



⇒ **Implication?**

- **Rising HPs will in long run (especially in superstar cities like London) lead to rising (intergenerational and wealth) inequality!**

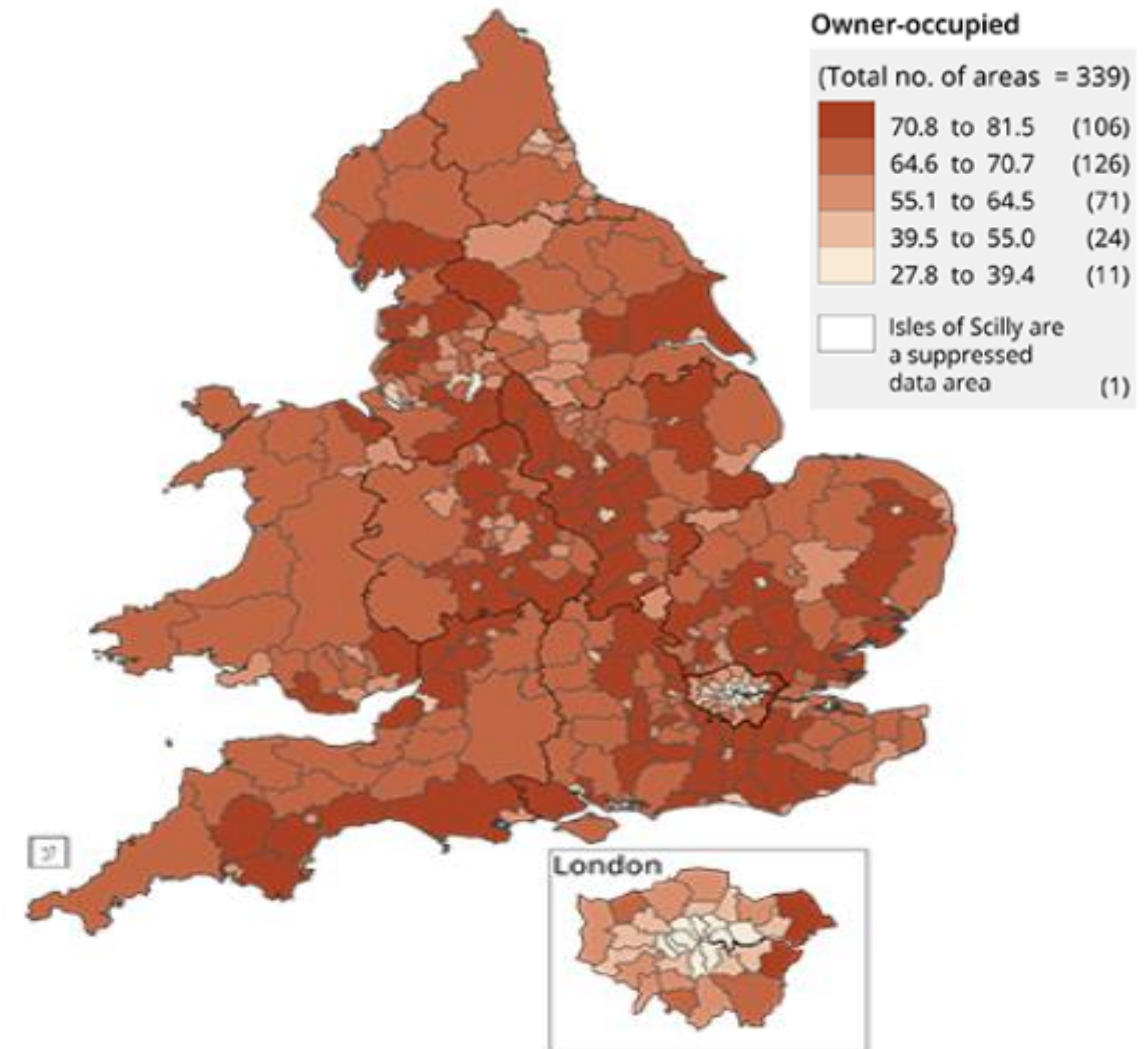
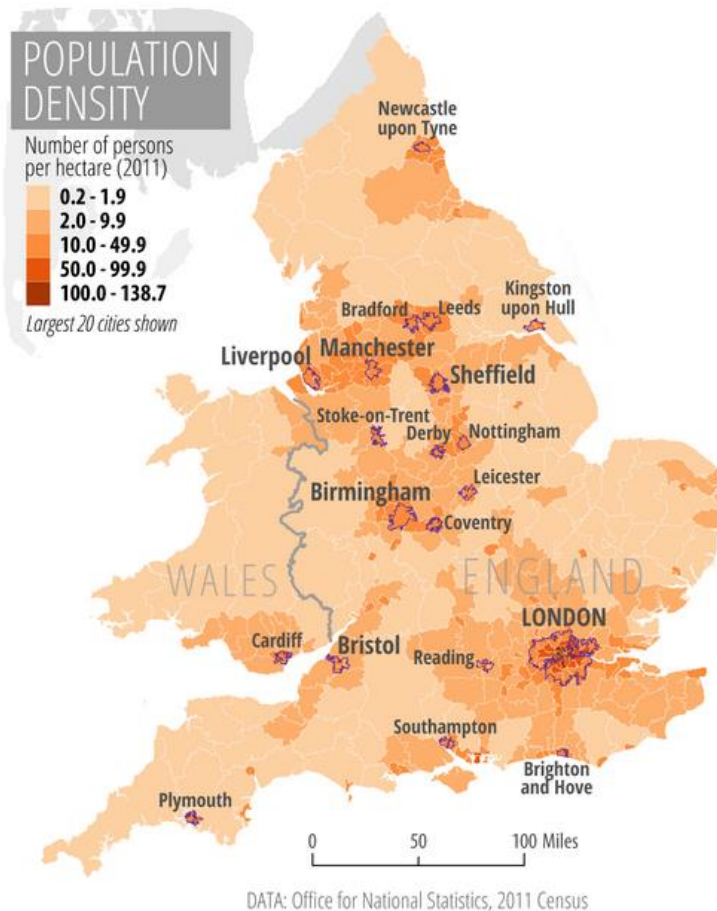
Source: Resolution Foundation (<https://www.resolutionfoundation.org/data/housing/>)

Stylised Fact 2

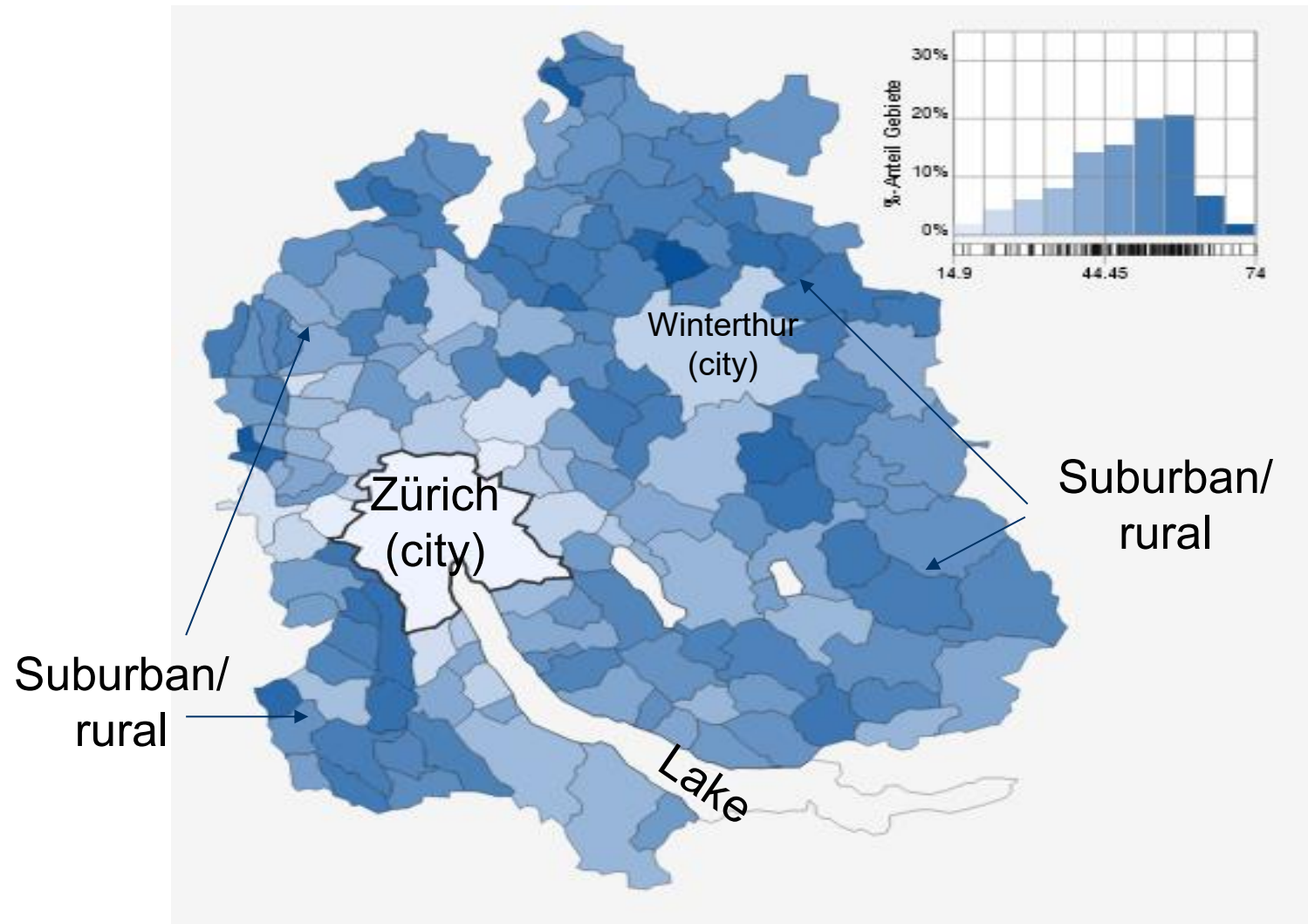
- Homeownership rates **vary strongly within** countries and even within regions & cities



Homeownership Rates Across Local Authorities in England & Wales (2018)



Homeownership Rates within Canton of Zurich, Switzerland (2021)



Source: Statistical Office Canton of Zurich (<https://www.zh.ch/de/politik-staat/statistik-daten/datenkatalog.html#/home>)

Stylized Fact 3

- Factors that determine **individual own-rent decisions** do not affect **aggregate homeownership rates** in same way





Stylized Fact 3



- Within country, higher income/wealth HHs much more likely to own in both countries **but** reverse picture when comparing aggregates:
 - Homeownership rate (2020):

Spain: 81.5%
Switzerland: 37.2%
 - GDP per capita (2021 in US\$):

Spain: \$30,103
Switzerland: \$91,992
 - GDP per capita PPP (2021, in US\$)

Spain: \$40,699
Switzerland: \$77,274

Sources: World Bank (GDP pc and GDP pc PPP), OECD Affordable Housing Database (homeownership rate)

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Motivation

- Homeownership has long been a goal of social policy in most countries
- In most countries owners of ‘principally owner-occupied dwellings’ (POODs) ...
 - ▶ Do not have to pay taxes on (imputed) rental income (in contrast to landlords)
 - ▶ Can deduct mortgage interest from taxes
 - ▶ Do not have to pay taxes on capital gains
 - ▶ Various other forms of subsidy, e.g., in UK: ‘Right to Buy’ or ‘Help to Buy’

Tax Policies Differ Across Europe

—Selected Countries

	Taxation of Imputed Rents	Deductibility of Mortgage Interest (Principal Unit Only*)
France	No	Yes* <i>(since 1998 only for repairs)</i>
Germany	No	No
Italy	No <i>(abolished in 2000)</i>	Yes*
Netherlands	Yes	Yes*
Spain	No <i>(abolished in 1999)</i>	Yes
Switzerland	Yes (till 2025) → No	Yes → Limited → Abolished
UK	No	No <i>(abolished in 2000)</i>

Sources: Hilber (2014, updated 2026)

Motivation—Cont.

- Policies are incredibly **costly**
 - ▶ Example: Mortgage Interest Deduction (MID) in US ~\$60B p.a. in foregone tax revenue until Tax Cuts & Jobs Act of 2017, now still ~\$25B
- Relevant research questions
 1. Are there any **social benefits** associated with homeownership that justify subsidies? Are there **social costs**?
 2. Are policies that favor homeownership **effective**? Do they increase homeownership attainment? Do they have unintended consequences?
 3. More generally: What factors **determine** homeownership attainment/rates?

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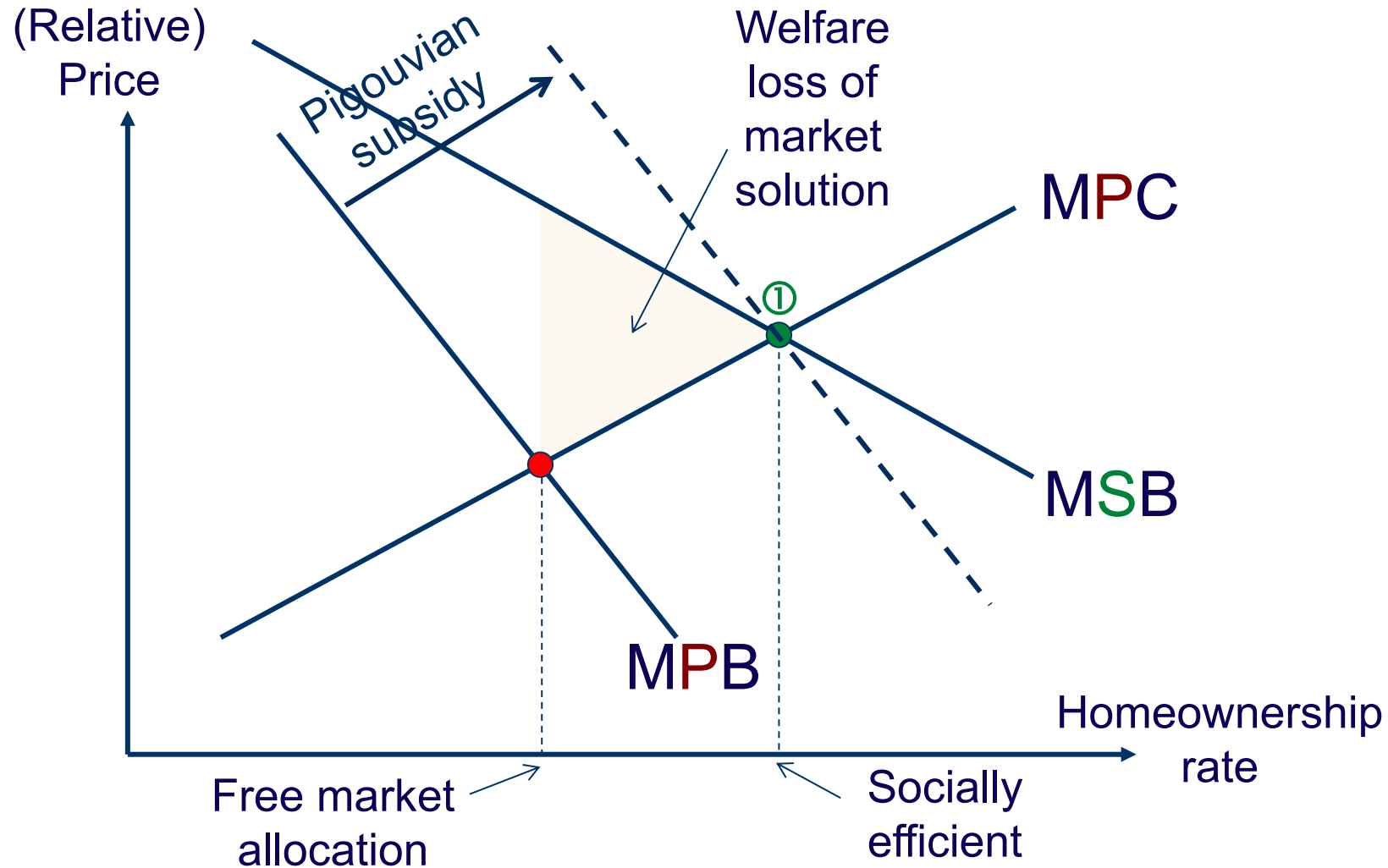
Welfare-Economic Considerations

- Homeownership subsidies may be justified IF
 1. 'Market failure' exists AND
 2. Positive externalities > negative externalities AND
 3. Policies are at least somewhat effective (no complete policy failure)
- Consider market failure first
 - ▶ Are there **social benefits** associated with homeownership?
 - ▶ Are there **social costs** associated with homeownership?

Excursus: Econ 101

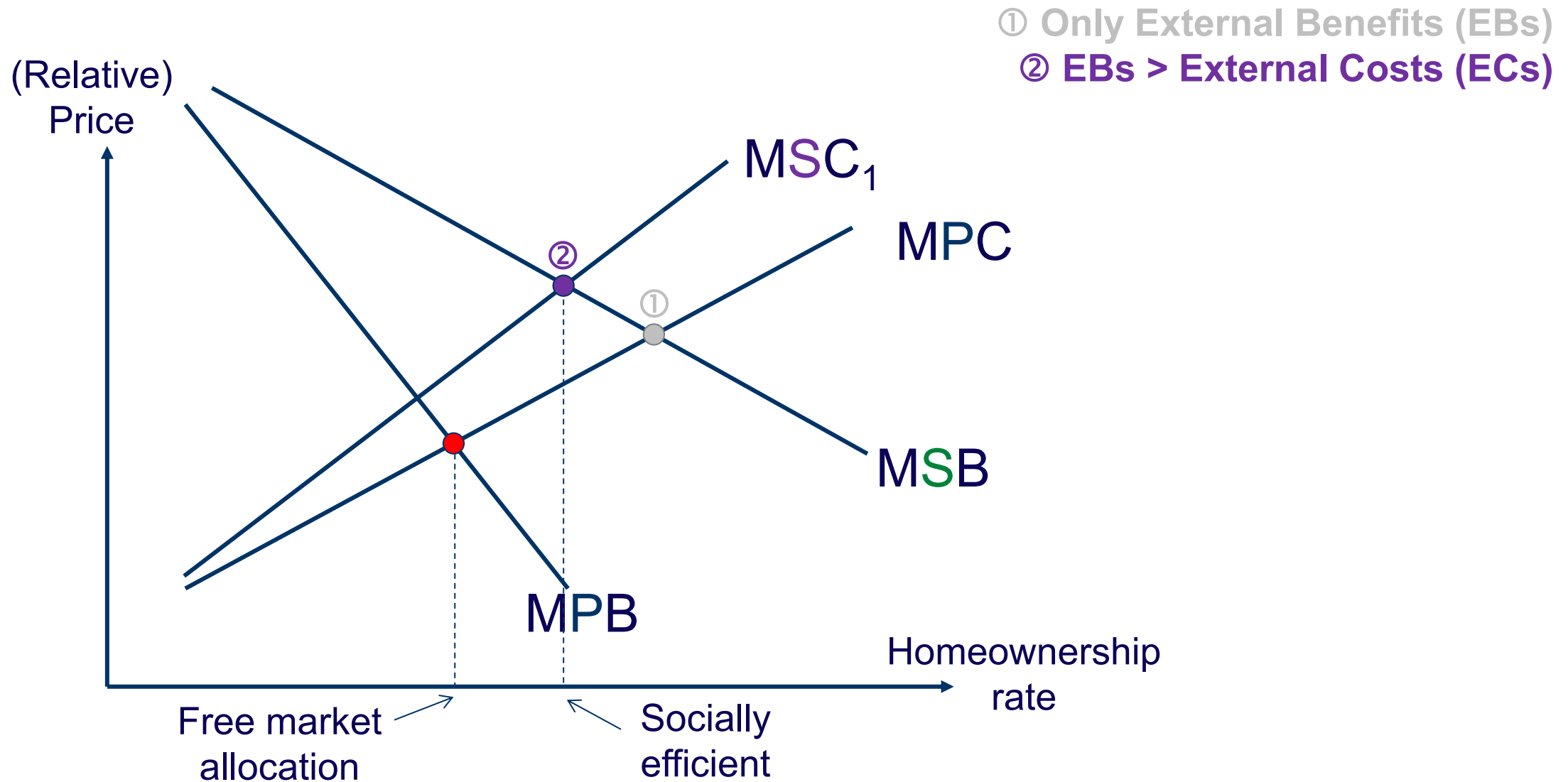
Positive vs. Negative Externalities

① Only External Benefits (EBs)



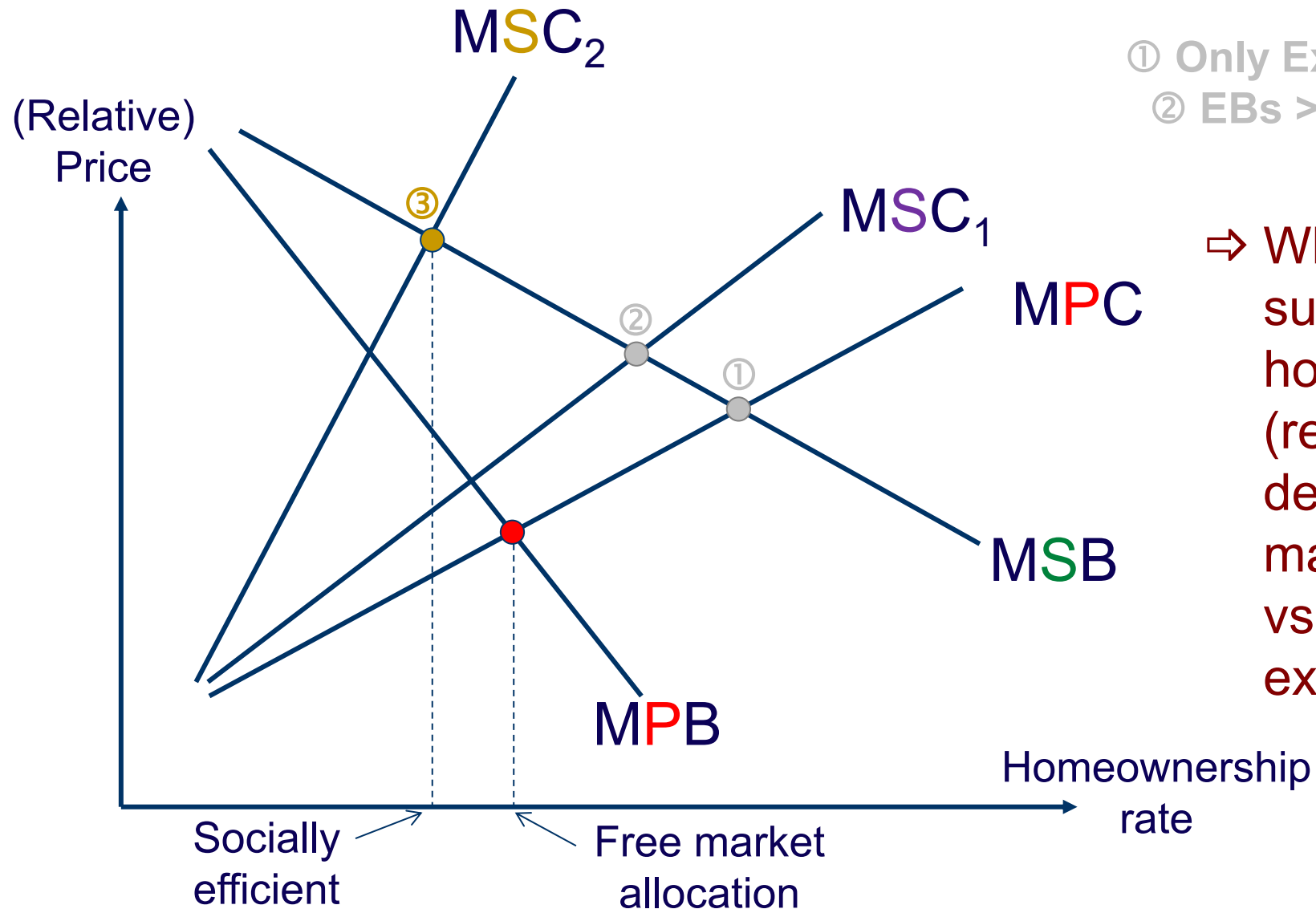
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Positive vs. Negative Externalities



Excursus: Econ 101

Positive vs. Negative Externalities



Are there Social Benefits?

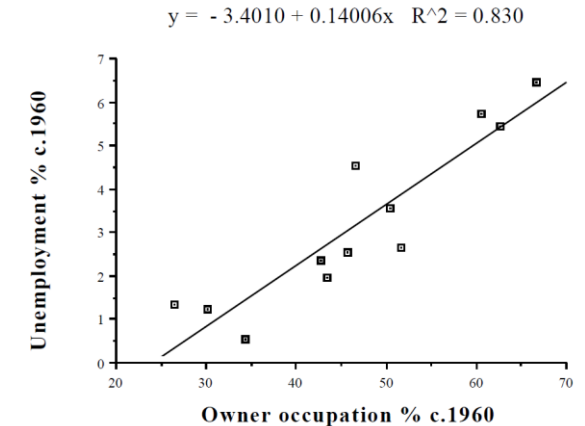
- Homeowners **treat housing stock more carefully** (Galster 1983)
 - ▶ ‘Renting to oneself’ prevents moral hazard
 - ▶ Neighbors benefit from better-maintained housing stock
- Homeownership leads to **more investment in social capital, more support for local public schools & enables better upbringing of children & reduces crime** (DiPasquale & Glaeser 1999; Dietz & Haurin 2003; Fischel 2001; Green & White 1996; Disney *et al.* 2023)
 - ▶ But positive links are mainly **confined to more developed physically places** (Hilber & Mayer 2009; Hilber 2010)

Are there Social Benefits? (cont.)

- Also: Homeownership leads to automated savings $\Rightarrow$ Less poverty amongst retirees (all else equal)
- $\Rightarrow$ *Some* evidence of **social benefits** (mainly in more urbanized places)

Are there Social Costs?

- Oswald (1996)
 - ▶ HOs are less mobile and hence more likely to be unemployed
- Munch *et al.* (2006) and many others
 - ▶ Confirm: Homeowners are indeed less mobile **but...**
 - ▶ *Individual* homeownership **does not cause** *individual* unemployment
 - ▶ **Why?** Homeownership improves chances of finding jobs locally (accept lower reservation wage)
 - ▶ Still: Individual homeownership might affect unemployment propensity *of others* via labor market mismatch



Are there Social Costs? (cont.)

- Blanchflower and Oswald (2013) provide new evidence for US of positive association b/w homeownership rate & unemployment rate
 - ▶ Reasons: (1) Homeownership lowers mobility, (2) increases commuting time & (3) leads to few new business
- Bracke *et al.* (2018) show that *leveraged* homeownership indeed reduces entrepreneurship propensity substantially (since leverage makes homeowners more risk-averse)

Are there Social Costs? (cont.)

- Homeowners may also be more likely to behave as NIMBYs (Fischel 2001; Hilber and Robert-Nicoud 2013; Greaney *et al.* 2026)
 - ▶ Homeownership ↑ → Restrictiveness of land use regulation ↑
→ Residential construction ↓

(‘Homevoters’ try to protect their asset values!)
- ⇒ **Some evidence of social costs too!** (But both external benefits & external costs **not** quantified)

Are the subsidies effective?

Theoretical considerations

- **If supply inelastic** – not per se clear
 - ▶ Present value of subsidy may be (fully) capitalized into higher house prices
 - ▶ This offsets positive effect of subsidy for first-time buyers
 - ▶ Moreover: Some down-payment-constrained would-be-buyers may no longer qualify for mortgage, and risk-averse may opt out (Hilber and Turner, 2014)
 - ▶ **Subsidy may have perverse effect of lowering homeownership attainment**
- **If supply elastic** – expect positive effect

Empirical Evidence

- In the United States (Hilber and Turner 2014; Sommer & Sullivan 2018)
 - ▶ In tightly regulated markets: **Strong adverse effect**
 - ▶ In less regulated markets: **Positive effect**
 - ▶ On aggregate: essentially **NO** effect
 - In the EU (Hilber 2014)
 - ▶ Abolition of taxation of imputed rents in Spain & Italy increases HOR by roughly 2%
 - ▶ Small but positive effect
- ⇒ **Subsidies largely ineffective**, but some positive effects in unregulated (mostly little urbanized) markets

Summing up...

1. Subsidies are incredibly costly to taxpayer, so needed significant positive externalities (PEs) to justify
 2. PEs likely exist but they are mainly confined to places with inelastic supply
 3. Negative externalities (NEs) likely also exist – not per se clear whether $PE > NE$
 4. In places with inelastic supply where PEs do exist, subsidy actually *lowers* (!) HO attainment
 5. In places where subsidy increases HO attainment, PEs are likely negligible (!)
- ⇒ **HO-subsidies like MID = ineffective policy to improve social welfare!**

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Determinants of Individual Tenure Choice (1/2)

- ⇒ Vast literature on determinants ranging back to 1980s...
- Relative user cost of owner-occupied housing (Rosen 1980; Hendershott 1980; Hendershott & Slemrod 1983; Poterba 1984)
- Household income & wealth (Rosen 1980; Poterba 1994; Gyourko *et al.* 1999)
- Liquidity & down-payment constraints (Linneman & Wachter 1989; Fuster & Zafar 2016)
- Discrimination & redlining (Ladd 1998; Dawkins 2005; Hilber & Liu 2008)
- Life cycle status (age, kids) (control in most tenure choice studies)

Determinants of Individual Tenure Choice (2/2)

- Transaction costs & expected duration in property (Haurin & Gill 2002)
- Income uncertainty, investment risk, neighbourhood specific risk & rent risk (Haurin 1991; Robst *et al.* 1999; Turner 2003; Hilber 2005; Sinai & Souleles 2005)
- Tax subsidies (Glaeser & Shapiro 2003; Bourassa & Yin 2008; Poterba & Sinai 2008; Hilber & Turner 2014; Sommer & Sullivan 2018)
- Property type & landlord production efficiency advantages (Linneman 1985)

What Factors Explain Spatial Differences in Homeownership Rates?

- Not necessarily same factors as individual tenure choice – Why?
- Example income
 - ▶ Within locality: Higher-income HHs are always more likely to afford owner-occupied homes/qualify for mortgage compared to lower-income HHs – they are also more likely to outbid them
 - ▶ Across locations: Higher incomes $\Rightarrow$ Higher housing demand $\Rightarrow$ capitalized into prices (& rents) $\Rightarrow$ Switzerland has much higher HPs (& rents) than Spain! City of Zurich has much higher house prices (& rents) than rural areas!

What Factors Explain Spatial Differences in HOR? (Hilber 2014)

- Look at restricted-use European Community Household Panel (ECHP)
 - ▶ **Panel** from 1994-2001 including roughly 60k HHs in 15 countries + geographical identifier
- Look at determinants of **aggregate homeownership rates** at regional level
- Include **rigorous controls**: Control for region FEs, country x year FEs, regional housing stock composition & regional demographic & socioeconomic composition

Results:

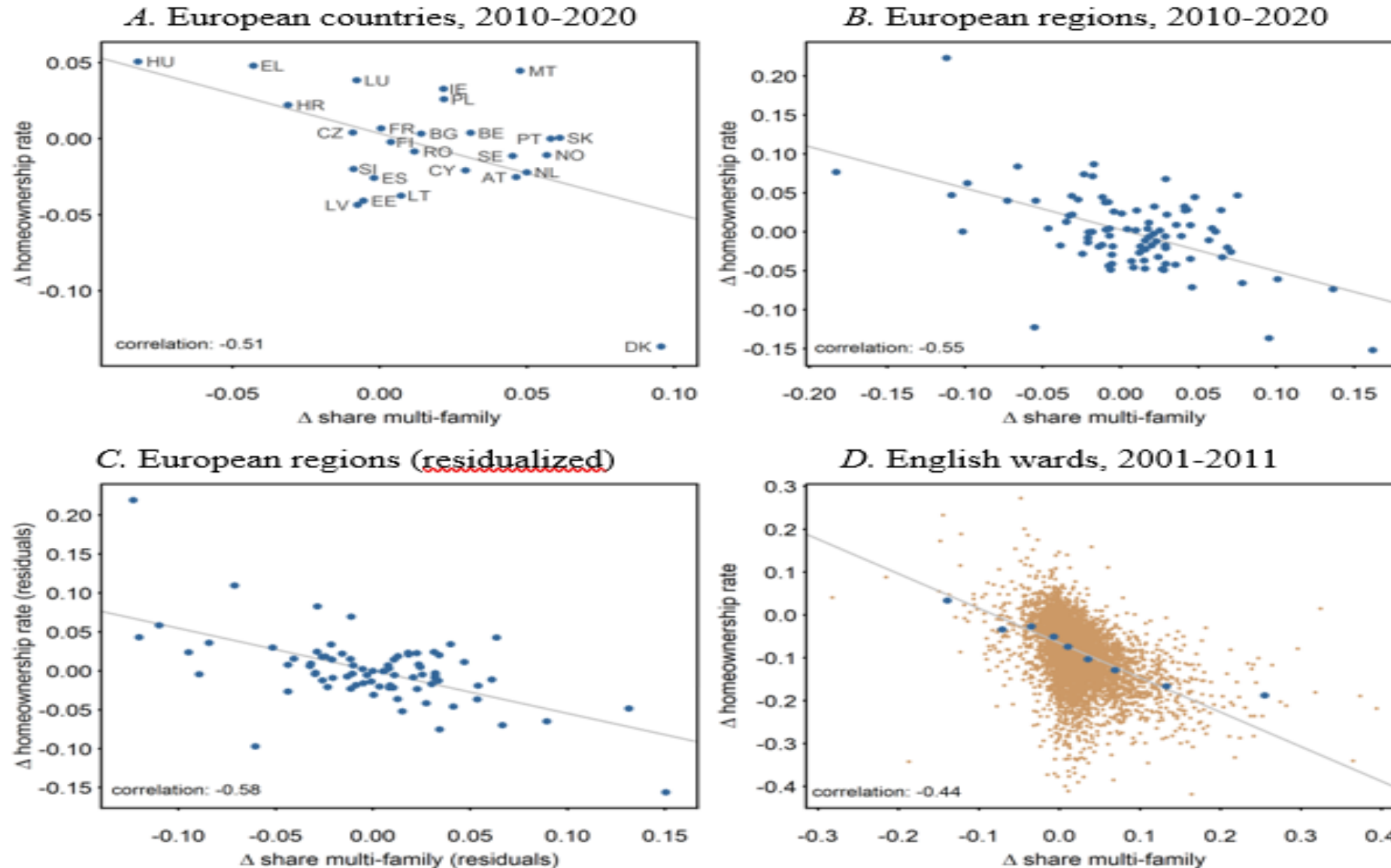
Region-level FE-model

Maximum Difference across Regions (NUTS 1&2)	Effect
HH age composition, 2000 Brandenburg (GER) least favorable → Algarve (POR) most favor.	+6.3% **
Av HH income (PPP-converted), 2000 Algarve (POR) lowest → Luxembourg (LUX) highest	[+2.1%]
Housing stock composition, 2000 Dublin (IRE) min MF → Berlin (GER) max MF	-30.6% **
Abolition of taxation of imputed rents Average of tax reforms in ITA & SP	+2.1% **
Share public rental units in region, 2000 Regions in GRE/LUX/SWE lowest → Mecklenburg (GER) highest	-13.6 **
%Young adults living with parents, 2000 Stockholm (SWE) lowest → Madeira (POR) highest	[-0.9%]

Further Evidence for Importance of Housing Stock Composition

- Hilber & Mense (2026)
 - ▶ Use soil conditions & insights from civil engineering literature to identify **causal effect** of housing type on homeownership in England
 - ▶ Key finding: **A 10 percentage points increase in share of multifamily housing causes homeownership rate to decrease by 8 percentage points!**
- Extension of analysis to 31 European countries
 - ▶ Share of multifamily housing is **negatively associated** with homeownership rate **in all 31 surveyed countries**, even when holding other determinants of homeownership constants

Visual Evidence for EU and English Words



Source: Hilber and Mense (2025)

Why Are Apartments (Flats) Less Likely to Be Owner-Occupied? (*Discuss*)

- High-rise buildings have higher **relative landlord production efficiency**
 - ▶ Easier for landlord to manage several units
 - ▶ Economies of scale (maintenance)
 - ▶ Coordination advantages
- ⇒ Would expect **high-rise buildings to be more likely to be renter-occupied than single-family houses**

Conclusions

- Homeownership has been associated with positive *and* negative externalities
 - ▶ Not clear whether subsidies justified on welfare economics grounds
- Subsidies like MID or HtB are likely ineffective in markets with inelastic long-run supply
- Homeownership attainment varies across locations mainly due to
 - ▶ Differences in housing stock (building types)
 - ▶ Differences in provision of public/social housing
- Sorting/segregation (by income, age etc.) and tax policies matter, but seemingly less important

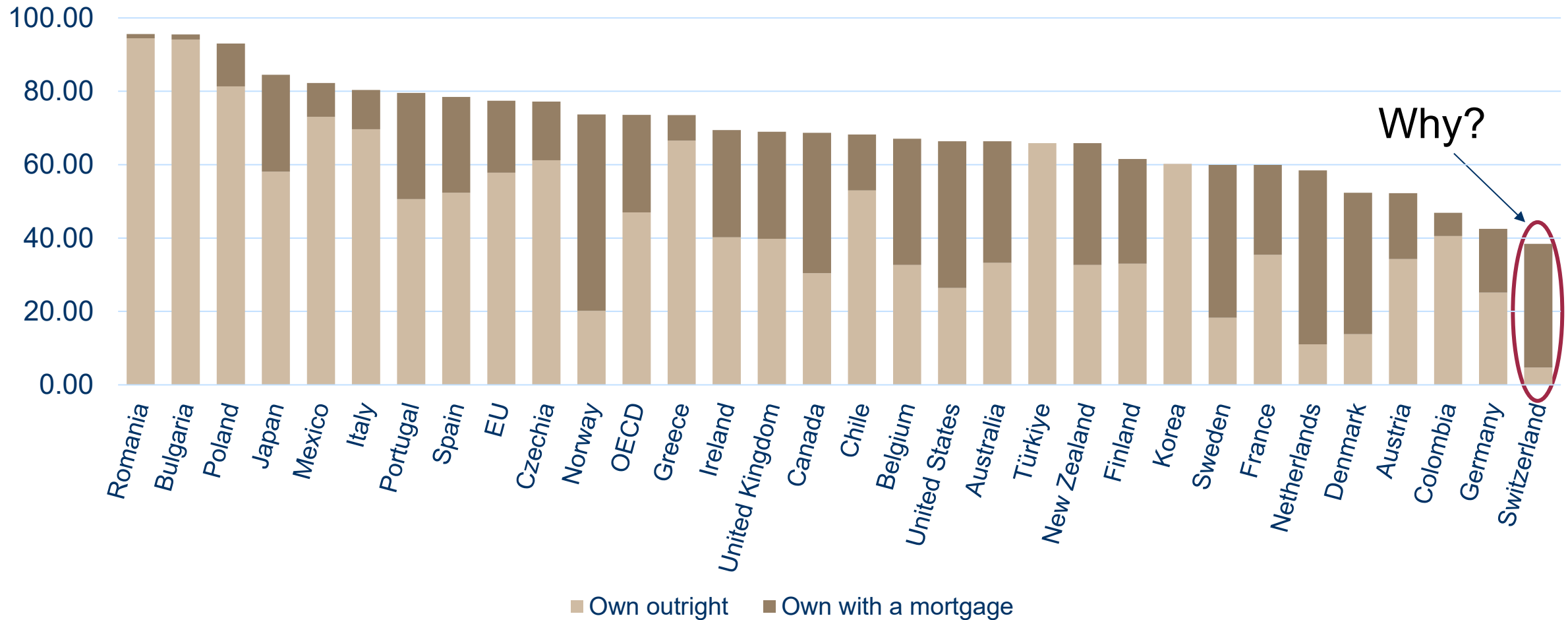
Preparations for *Block III—Seminar 6* (Week 10)

- **Topic:** Should governments subsidize homeownership?
- **Seminar format:** Key paper presentations (see handout on Moodle)
- **Papers to be discussed:** DiPasquale and Glaeser (1999) and Hilber and Turner (2014)
- **Tasks for all students:** Read key papers and prepare questions for presenters

Additional slides

Homeownership with and without mortgages – OECD countries (2024)

◀ Stylized fact 1



Source: OECD Affordable Housing Database

(<https://www.oecd.org/content/oecd/en/data/datasets/oecd-affordable-housing-database.html>)